

Instructions for the end of term project.

Please read the whole document carefully. I appreciate that it is long, but I hope it answers many of the questions you may have about what to do.

Motivation

Group work

Aim:

Your group works for a statistical consultancy company (you choose the name) . The company has been contacted by the Department for Work and Pensions (DWP) and the Department for Education (DfE) to analyse data from a longitudinal cohort study they commissioned. The DWP and the DfE are particularly interested in understanding what life factors are associated with higher incomes at 25 years of age and what, if any, effect disabilities, learning difficulties, ethnicity and sex have on income. For a bonus, they are also interested in understanding whether there is a pattern to the missing entries and whether in future studies they should make more effort to sample certain types of people.

Output:

The government departments have resident quantitative staff who understand statistics and will want a clear and detailed description of how the analysis was conducted and relevant results. However, the information also needs to be relayed to non-statisticians working in DWP and DfE who can then make recommendations to policy makers in the form of a more accessible piece of work. It is unlikely that your consultancy company will be asked to do any more work for government if either one of your outputs is unsatisfactory. Presentation is important, but clarity is even more important for both outputs.

Details:

The study follows the lives of around 16,000 people in England born in 1989-90 and began in 2004 when the cohort members were aged 14, with an original sample of 15,770 people. Cohort members were surveyed annually until 2010, and the next sweep after this was when they were aged 25, in 2015-16.

The data file you are given (INC.csv) is a subset of the data on which to run your initial analysis. The data are in the format in which they were originally collected, including the names given by the data collectors which are not very intuitive. For example, all variables start with WX where X is a number from 1-8 representing the wave in which the information was collected. The data you have been given is compiled from waves 1 to 8, excluding waves 3 and 7. The data dictionary and translator are on the Moodle page.

The translator has the year the variable was measured, the school year of the young person (where relevant), the age of the young person, the name of the variable, its type, and a short description. The exact details of its coding are in the data dictionary. If you want to know the original data dictionaries, please let me know and I can send you a copy. You will have to search the data dictionary for your variables. Spend some time familiarising yourself with the data. Please post in the forum if you have questions or are struggling.

The outcome of interest is called W8DINCW which is an income related outcome in Wave 8. The data have been reduced so there are no missing data in the outcome.

Individual work

Aim:

For the Individual project, your consultancy have been commissioned by NHS and NHS Scotland to analyse data from a survey run in England and Scotland. This is a subset of a larger dataset that looks at how health and gambling outcomes are associated with each other and underlying socio-economic, demographic and life-style factors.

Your boss has asked you to look at the outcome of whether someone is a problem gambler (PROBGAM), and has also asked if you can focus on **one of** health, socio-economic or life-style factors. This does not mean that you should ignore other predictors (as these may be highly relevant confounders) but that you should focus on one set of these and e.g. possible interactions etc.

Output:

As with the group project, there will be a technical report and a non-technical report. The format of the non-technical report for government officials is a *one page* summary of your main findings. This can include some plots, but make sure they are relevant.

Details:

For the introduction, you don't need to research the dataset, but you must motivate your research question (why health, socio-economic or life-style factors), and briefly outline the main conclusions.

All the details

The **end of term project (EOT) is worth overall 90%** of your final grade. (10% was allocated to the reading week project). There are 2 components to the EOT:

- I. A group project using **multiple linear regression** and the Next Steps data set – worth 55% of your final grade. This has two sub-components
 - A. A report for which you will be expected to use multiple linear regression and all the associated techniques that we have covered in the course to explore, analyse and interpret the data.
 - B. Evidence of group collaboration in the form of:
 1. Create a Google Sheet that will keep track of your meetings and that you will submit together with a common report. These are called the meeting logs
 2. Each group member will write an individual evaluation of their contribution in each area of the project and a statement on how the group collaborated.

Failure to submit the group collaboration component will result in a 0 mark. Further details of these components are given below.
- II. An individual project using **logistic regression** on gambling data.

You will be expected to use logistic regression and the associated techniques that we have covered in the course to explore, analyse and interpret the data. This is individual work, I will be checking both the text and the R code for plagiarism.

The deadline for submission via Moodle for all pieces of work is **Thursday May 15th 2025 at 4 pm**.

Submitting work

There will be three submission portals. One for the group project and meetings log, one for the collaboration evaluation document, and one for the individual project. **Each group will submit one report to the group project portal and each student will submit one report to the individual project portal and one collaboration document to the collaboration portal.** For each portal the submissions are:

1. A copy of a reports/collaboration statement as a **pdf file**. Hide the code if you use RMarkdown. For the group project include the candidate numbers of all group members in the cover page (NOT YOUR NAMES).
2. A copy of their **R code as a .txt file**. Moodle doesn't accept R files or Rmd code if they used Rmarkdown for the group project.
3. FOR THE GROUP PROJECT ONLY: a copy of your meetings log as a csv or excel file.

For the group project : name your project "ST211_group" followed by the group number. I want a cover page to contain a list of the candidate numbers of your group – NOT your names . At the end of the project, you should append the meeting logs (more about these later).

For the individual project name of your uploaded files should be "ST211_individual" followed by your candidate number. Please do not include your name.

For the collaboration evaluation the name of your uploaded files should be "ST211_collaboration" followed by your *group* number. Please do not include your name or candidate number.

Word count: 2000-2500 to the group project and 1000-1500 to the individual work. Please report the word count of both projects, on the cover page. The word count does not include the R code or the appendix (details later).

I will pass the reports through Turn-it-in to ensure that there is no plagiarism.

Contents of the report

You received exhaustive advice on how to write a report for the reading week project in the guidelines for that project as well as the feedback sessions. The word count is longer so you may write more for this report in each section than you wrote for the reading week report. I will not be checking for individual section word counts.

For each outcome you need to formulate a **research question**. This should be clearly stated in your introduction and lay interpretation. I will look favorably on a well formulated research question, however a general research question – how do the predictors affect the outcome, is also acceptable.

Each report should have a technical part and a non-statistician (lay) part as with the reading week report.

For the MLR group reports:

The technical reports

The **technical** report should be a clear and concise description of the analysis with focus on what methods were used, why they were used, how you used the results of these methods to make decisions about the next model etc. It should not contain any R code as any statistician, even those who do not use R, should be able to read it.

I will be assessing how well you understand and perform the analysis. I am less interested in the quality of the model – although it also needs to be considered. In fact, if the best model you can obtain is still a bad model, it is important that you state this clearly. You are using linear regression, and while it is a great technique, it is not always adequate to describe complex data sets. You should not try and persuade me that the model is excellent when in fact it isn't.

There should be a brief technical interpretation of the results at the end of the report which compares different models e.g. with/without interaction, outliers, transformations, missing data.

There are **missing data** in the data sets. These can be missing, refusal, don't knows etc. and are typically coded as negative numbers. There may also be some NAs where there has been drop-out between subsequent waves. I have already removed individuals who are not in Wave 8. Due to drop-out the initial sample has gone reduced significantly from 15,770. Some of the continuous predictors have a lot of missing data. You may want to consider excluding these altogether, but if you do, make sure you justify your choices.

I am suggesting two approaches below. The second approach can lead to additional marks for this part, but only if done well.

For the first approach, I expect you do the following: If a categorical predictor is missing less than 10% of its values, you can remove those and just say that you removed them. The problem with that is that you may end up with very few points if the 10% are different for each predictor. So use this rule with caution. A lot of missing is going on in the continuous predictors, some of which have over 30% missing. For these, I am happy for you to remove these predictors from a final model to see if they change the regression substantially, provided of course that they are significant.

For categorical predictors with more than 10% missing: Replace all the missing values with NA and then proceed with your regression. The final regression will then be a complete case analysis on the predictors in your model.

For the second approach, start off with the method shown above and then

1. Replace all the missing in the continuous predictors with NA and with a "missing level" in the categorical predictors and proceed with your regression. In the final model, replace the missing with NA and compare the results
2. For some interesting predictors (you must justify why they are interesting) explore in more detail the effect of the missingness. Is there a pattern, is missingness predictive of the outcome? What consequences could this have on the bias.
3. Do NOT use multiple imputation.

Just attempting the second approach and doing it badly will not result in additional marks. In fact, if this is done badly, or incorrectly, this may result in negative marks. So think about it. I will not be telling you how many more marks it is worth.

Bear in mind that these are not techniques you would normally use in real contexts (generally you would use multiple imputation or more complex models for informative missingness), however I hope that by exploring these different options, you may get an insight into how missing data can affect results.

What to include

In this report you will include discussions of **outliers, transformations, interactions, collinearity and cross validation addition to what was covered in the reading week project**. Refer to the *Guide to Model Building and Assessment* at the back of the course notes and on the Moodle Resources section. If you cover everything that is listed there and in approximately the order suggested there you will do well. **I expect you to attempt all the techniques we discussed in the class and to check for all possible problems** (collinearity, outliers etc.). If these are not a concern, then it is sufficient to mention that you attempted them and that the results were not interesting. Be specific, if you tried an interaction, say which variables you interacted *and why*. If you did an outlier analysis but found none then say, we looked at standardised residuals over 3, cooks distance, DFFITS and leverage values but found no outlying values and therefore did not pursue this further.

I may check your code to verify that you have indeed performed these analyses. Bear in mind that I have also analysed the data and that I will know if these issues are present.

The appendix

As with the Reading Week project, you will probably merge the levels of some of your categorical predictors. This information should go in the **appendix**. Any other data manipulations should also be listed and briefly justified in the appendix. Do not include in the appendix any information about the analysis itself as I will not count it.

The lay report

The **lay** report is a short piece of writing that is aimed at non-statisticians. The format of the lay report is up to you. Again I suggest two approaches. The first is to write this up as a simple report, aimed at a non-technical audience, specifically the government officials.

For extra marks, a very well executed alternative format can be considered. For example, an annotated presentation, a short video presentation whatever you think is appropriate. Feel free to ask me before you do this and as with the missingness, doing this badly may well result in fewer marks than a simple report.

Please remember **not to use judgmental or prejudiced language** in your lay report. The lay report should contain **no technical language e.g no fractions, significance, diagnostics etc.** Any predictors need to be explained in words and not referred to by their names in the dataset. **It is also not a tutorial about regression.** The idea is to take the research question(s), data and results from the analysis and make them accessible. Avoid plots unless you feel they explain more than they need to be explained.

The code

You are required to upload the code for the report. The code must run on my machine with the original data. I will NOT accept datasets that have been manipulated in excel as uploads. All data manipulation must be done in R (not python or any other language). I am happy to provide some guidance on this on

the forums. Data manipulation will involve things you have hopefully done already in the RW project such as:

1. replacing the missing values (typically coded as -ve numbers) by either NA or “missing”,
2. merging levels of categorical predictors,

as well as things you haven’t done with real data but that we have covered such as

1. transforming an outcome using a log transformation
2. turning a continuous variable into a binary variable

I am more than happy to answer coding questions as I detail below.

You do not get awarded points for good code, you get docked points 20% for code that does not run.

Group work: meetings log and evaluations

A list of rules for good group work can be found on the Moodle page. I will take very seriously any allegations of bullying or harassment of one or more group members. I have a no-tolerance policy and people involved in bullying or harassment may have to face disciplinary procedures.

Here I will talk very briefly about **work allocation**, the **meetings log** and the **evaluation report**.

- 1) **Each group will set up a Google sheet which they will share with me where I can monitor group meetings and participation.** This document will be updated every time there is a meeting. A template can be found at the end of this section.
- 2) *I will expect to see a meeting with everyone participating before the end of the term and 3 additional meeting scheduled that the whole group has agreed to. More meetings are recommended and should be documented.*
- 3) All of the group members should become familiar with the dataset by the time you have the first meeting.
- 4) You can then discuss how to proceed and allocate tasks to different group members. I would recommend allocating the work so that everyone is doing some exploratory data analysis, some analysis, some writing of the technical report and the lay report (this is essential for the evaluations of the contribution). Whatever the work allocation is, you need to check each other’s work. E.g. compare codes, combine analyses that are for the same dataset, make decisions about what to do next and then divide the tasks. For example, someone does the outliers, someone looks at interactions and then you can meet again and swap codes to check for errors etc. Then decide who writes up what.
- 5) It is a good idea to set up all future meetings in advance. E.g. every Monday at 11 on Zoom.
- 6) Everyone should do some of the coding and some of the writing up. If you allocate all the work by perceived skills, some people may end up doing most of the work and if they get it wrong, the whole group will pay the price. Equally, it will mean that those who do all the work will resent those that don’t and will come to me to ask for the group to be broken up or for people they perceive to have done less to be excluded. Finally, no-one will challenge themselves to do something they aren’t great at and won’t learn anything new.
- 7) You could divide the group in two and each part can analyse one group of predictors and then you swap code/notes and make sure the two analyses are consistent.

- 8) You should be honest from the beginning about your **time availability** so that the group can work around it. For example, if someone has less time at the beginning they can do more work later on.
- 9) Going to regular meetings ensures that even if you are not working on something all the time you are aware of what is going on.
- 10) **All decisions need to be made and agreed to as a group. Communication is key**
- 11) This is no different from how you will work once you leave university. Whether you get along with colleagues or not, you will have to be able to work with them. Your employers will not be impressed you if you do not contribute and will also expect some level of teaching/learning among colleagues.

The process should be documented in a **meeting log on the Google Sheet**. This is explained below. At each of the meetings – pre-scheduled or scheduled as needed, you will:

1. log the attendance,
2. review the task allocation from the previous meeting,
3. discuss the results and how to move forward,
4. allocate new tasks,
5. set a new meeting date if you would like an additional meeting or need to reschedule of meetings already planned.

You will hand in the meeting logs with your group report as an additional file. Even though I can already see them in the Google sheets, they need to be on record as they are summative. . One entry in the logs could look like this:

Date	Attending/Apologies	Tasks from last meeting	Tasks for next meeting	Other comments
10/04/22	Jim, Yining, Amy, Hassan	Jim & Yining: Check outliers Amy: Write up analysis so far Hassan: check residual assumptions and diagnostics	Etc.	J&Y didn't finish their task from last week so will catch up as well as completing this week's task Amy wasn't well so she was only able to write ...

Contribution evaluations

Every member of the group will submit their evaluation to Moodle in the portal set up especially for this. You will use call these files ST211_group_number_evaluation.docx. The format will be as follows:

1. A sentence or two about your contribution to each of the main aspects of the report: The exploratory data analysis, the modelling (this includes decisions about what variables to keep, what missingness approach to take etc), the coding, the write up and the lay report.
2. A table with your self evaluated contribution, in % to each part of the report.
3. Optional: If you feel someone did not contribute or did a very good job. This may not be taken into account as you should have informed me of this before the submission of the report.

An example may look like this:

EDA: I did this and that

Modelling: I did this and that. I did not do the other

Coding: etc.

Write-up: etc.

Lay report: etc.

Component	Contribution
EDA	x%
Modelling	x%
Coding	x%
Write up	x%
Lay report	x%

Overall the group collaborated well, however xxxx. What really worked well was xxx.

Group conflicts

It can sometimes happen that groups do not work well together. If this is the case with your group you should first of all try and work something out among yourselves. If this does not work, let me know **as soon as possible**. If you do not let me know within 20 days of the deadline, I will be unable to help and the work that is handed in will be treated as the work of the whole group.

When there are conflicts, this typically arises because one or more group members are perceived as not participating sufficiently in the work. If this is the case and you are unable to solve this issue, please let me know as soon as possible. The meetings log will then provide evidence. I will not look kindly on groups who intentionally exclude one or more members, for example by saying that they are less capable, are not available when the rest of the group wants to meet etc. If your group decides to split up (for the reasons listed above), I may apply a penalty.

If you are not able to complete the work you have been allocated by the group, please let me and your group know as soon as possible. There are options such as deferrals or extensions that we can discuss. ST211 is a crucial course for many of the 3rd year options and not engaging fully in the assessment will have negative consequences later.

For the Logistic regression Individual reports

Follow the same structure for this report as you have with the linear regression report/reading week report. Bear in mind that we have covered fewer techniques in logistic regression than in linear regression and that there are fewer diagnostics. I expect you to check multicollinearity, and to do confusion tables (predicted vs observed) but no outliers, interactions or cross-validation (unless you want to do interactions and cross-validation as these can be done with the techniques we have covered)

In these data, missing values are indicated by an NA. For this report, *you do not need to deal with the missing data, just leave them as NA.*

You will need to run an exploratory analysis using for boxplots or barplots to understand whether there are any associations between the various predictors and the binary outcome. Please only document these in the exploratory data analysis, as with the MLR, I do not need to see the plots.

More details on the data are given in the data dictionary on Moodle. Please make sure you read this carefully and research any terms you don't understand and ask me on Moodle if you are still struggling.

Some points to consider are:

- Many predictors have many levels, you will have to do some merging for some of them, possibly even the predictors of main interest.
- If you run a regression with all the predictors and all their levels, most predictors won't be significant, possibly even those of interest, so it's worth starting with a smaller model that includes your predictor of interest. So forward selection instead of backward elimination.
- For this report only, I want you to keep the at least some of the predictors in your set of interest in the regression, even if they are not significant.
- All variables are coded as numbers but only some are actually numeric (see the data dictionary)
- Note about VIF for categorical predictors. When you use the VIF function and there are categorical predictors your output is slightly different (GVIF is used). Interpret this in the same way as the VIF.

Note: The individual project is both harder (logistic regressions are a bit more difficult to interpret and you will be less familiar with them) and easier (they have fewer diagnostics, no interactions or outlier analyses are required).

If you have any questions, feel free to post on the forum.

Questions

The projects are a major undertaking so I will be available to answer questions on the forum over the break although I will also take some time off. I will also arrange two one-hour online Q&A sessions.

I will answer:

1. Questions about how to code something.

2. Questions about the diagnostics (e.g. If the VIF > 10, should I remove the variable?)
3. Questions about the instructions
4. Questions to clarify definitions of the outcomes/predictors
5. What do you mean by health/socio-economic/life-style factors?

I will **not** answer:

1. Specific questions about your analysis: (e.g. should I use the predictor xxxx in the regression with yyyy as the outcome?)
2. Is my regression/result/conclusion correct?

Feedback

I will be releasing individual feedback in the form of a set of bullet point comments within 6 weeks of the deadline.